Isaac Abraham

732-209-1375 | isaac.abraham@rice.edu | linkedin.com/in/-isaacabraham | isaacjabraham.github.io

EDUCATION

Rice University, George R. Brown School of Engineering, Houston, TX *Expected May 2029*
B.S. Electrical and Computer Engineering

EXPERIENCE

Researcher, Experiential Pixels Lab, Rice University *Houston, TX | Sep 2026 – Current*
- Synchronize and calibrate the lasers and video cameras on the Volcap volumetric capture rig through a browser-based web app.
- Develop components of the rig's distributed capture system, including Android camera control and network configuration.

Electrical Subteam Member, Rice Racing (SAE) *Houston, TX | Sep 2026 – Current*
- Wire the race car's electrical system, connecting the electronic control unit (ECU), sensors, and other vehicle electronics.
- Gain hands-on experience with automotive electrical components, including ECU operation and sensor integration.

Researcher, Computer-Human Interaction Laboratory, Rice University *Houston, TX | Jan 2026 – May 2026*
- Developed a machine learning algorithm to parse and classify PDF voting ballot components for faster, more accurate elections.
- Implemented object detection techniques, including circularity-based formulas and size normalization, to identify and classify voting ovals and text fields within PDF documents.

Extern, Cisco *New York City, NY | Jun 2024 – Jul 2024*
- Led a research project on mitigating ransomware threats in the manufacturing sector, leveraging Cisco networking and security products alongside generative AI tools.
- Collaborated with Cisco team members to scope IT engineering challenges, determined the best application of Splunk AI cybersecurity solutions, and presented the resulting strategies to leadership.

Researcher, Inspirit AI Scholars Research Project *Virtual | Summer 2024*
- Built AI models in Python, including a convolutional neural network for tumor diagnosis and logistic regression for car valuation.
- Designed a recidivism risk assessment algorithm that reduces the bias found in existing judicial models, and won Best Presentation for work on the ethical drawbacks of biased AI.

Curriculum Manager, Student Alliance with Refugees & Immigrants *Princeton, NJ | Summer 2024*
- Developed and taught two 3-month computer science curricula (programming, hardware/software, AI) for the nonprofit TutrMePrep, serving Afghan refugees in the Princeton area.

PROJECTS & RESEARCH

Deepfake Video Detection (Neural Network) *Summer 2026*
- Paired a pretrained EfficientNet encoder with a temporal transformer in PyTorch to flag AI-generated videos, reaching 74% real-video accuracy and 62% deepfake detection on unseen actors in Google and Jigsaw's Deepfake Detection dataset.
- Engineered a preprocessing pipeline for frame sampling, resizing, ImageNet normalization, and augmentation (horizontal flip, brightness jitter) across ~60,000 frames from 1,800+ videos, using actor-level train/test splits so the classifier learns video patterns rather than actor identity.
- Diagnosed a generalization gap (92% accuracy in-dataset vs. 24% on an unseen generation tool) and a class imbalance that flagged 78% of real videos as fake, then reported real and fake accuracy separately instead of one blended score.

Human vs. LLM Performance in Data Extraction (Research Paper) *Dec 2025*
- Authored an IEEE-style research paper benchmarking the data-extraction accuracy of five large language models (Gemini 2.5 Pro, Claude Sonnet 4.5, GPT-4o, Grok, DeepSeek) against manual data extraction with WebPlotDigitizer on a complex graph.
- Computed percent error across ~350 data points, finding LLMs unreliable for precise extraction (best-case 9.71% average error) and identifying overlapping visual elements as the primary reason for failure.

AI Ethics & Policy Writing *2024 – 2025*
- Authored a paper analyzing three ethical risks of AI (biased training data, moral dilemmas in autonomous vehicles, and black-box opacity) and proposing regulation based on the ACM's six core principles for generative AI.
- Authored an IEEE-style paper arguing that as agentic AI automates white-collar work, the human skills AI lacks (empathy, judgment, and creativity, per MIT's EPOCH framework) will become more valuable, so education should prioritize them.

SKILLS & INTERESTS

Programming: Python, Java, JavaScript, MATLAB
Hardware: Verilog, FPGA design (Vivado), LTspice, KiCad, breadboard prototyping
Tools: PyTorch, Splunk, Xcalibur, Word, Excel, Outlook
Languages: Spanish (Proficient), German (Beginner)
Interests: Piano, AI & Ethics, Swimming, Writing